



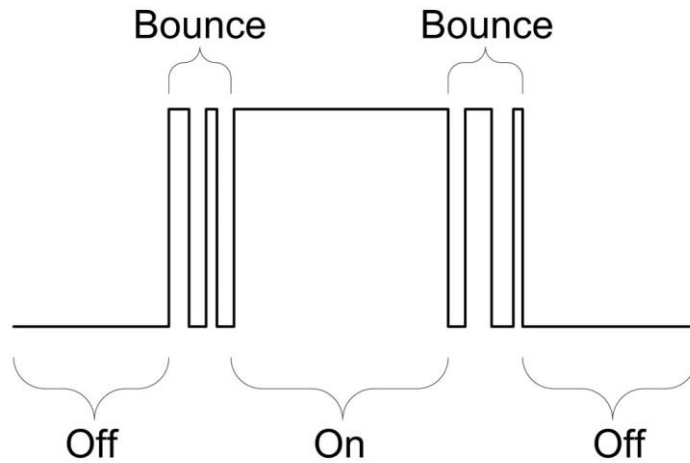
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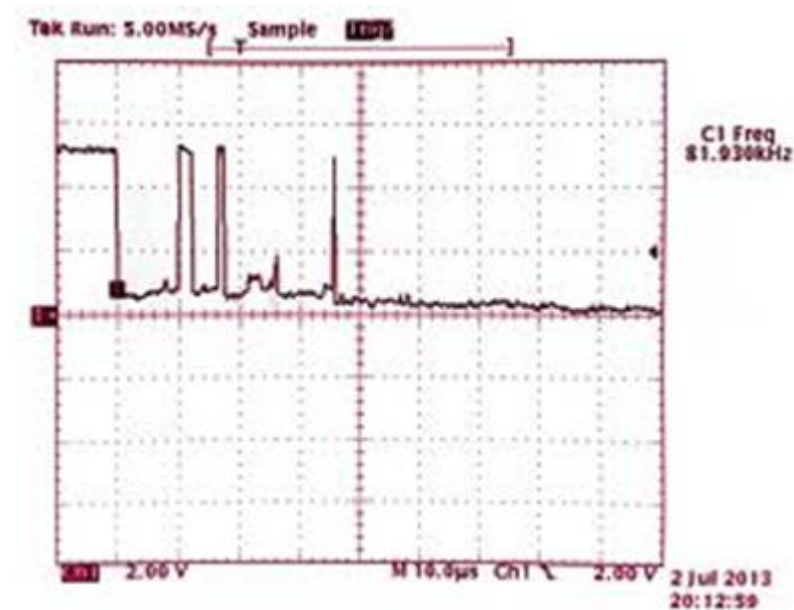
FPGA Design Using VHDL

Module 6H: Debouncing Hints

Button Debouncing 1



source: protostck.com



source: hifiengineering.blogspot.com

Button Debouncing 2

- Want to design a hardware component that eliminates bouncing
- Assume that you press the button for 0.1 seconds or longer
- One method to eliminate bouncing
 - Use a counter that increments as long as you press the button
 - If you hold the button for more than 10,000,000 counts, then assert a db signal high, otherwise keep db signal low

```
if(pb0 = '1')
    if(counter < 100000000)
        increment counter
    else
        pb0db <= '1'
else
    pb0db <= '0'
    counter <= 0's
```

Button Debouncing... starter code

```
process (clk,reset)
Begin
    if (reset='1') then
        pb0_db <= '0';
        db_cntr <= (others=>'0');
    elseif (rising_edge(clk)) then
        if pb0 = '1' then
            if db_cntr < 100000000 then
                db_cntr <= db_cntr + 1;
            else
                pb0_db <= '1';
            end if;
        else
            pb0_db <= '0';
            db_cntr <= (others=>'0');
        end if;
    end if;
End process;
```

Button Debouncing Continued

